

Agent Architectures

From Classical Agent Architectures to LLM-Based Agentic AI



Outline

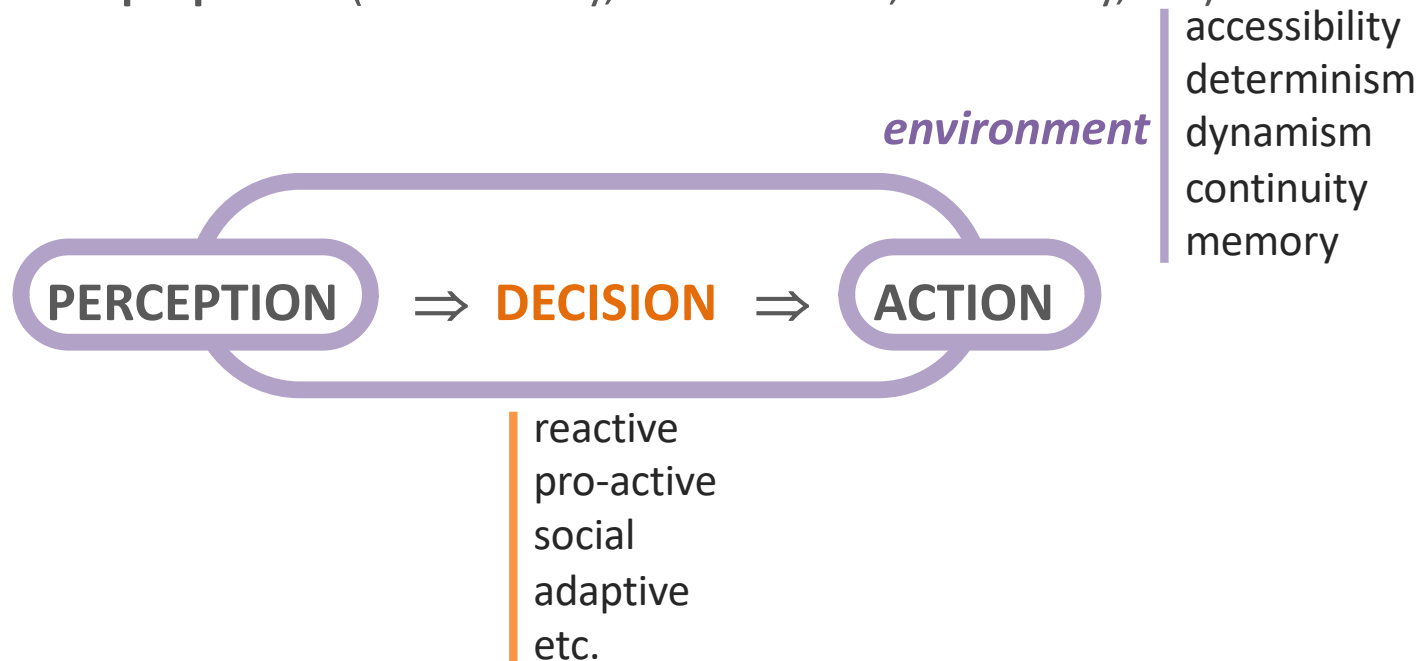
- Classical agent architectures
- BDI reasoning
- Reactive & hybrid systems
- Modern LLM-based agents
- Agent frameworks (LangGraph, AutoGen, CrewAI, etc.)



Recalling concepts

AGENT & ENVIRONMENT

- Sense – Decide - Act
- **Agent properties** (reactive, pro-active, social, adaptive, etc.)
- **Social Ability** (cooperation, coordination, negotiation)
- **Environment properties** (accessibility, determinism, continuity, etc)



What is an Agent Architecture?

- Structure describing **agent behavior**
- Defines **internal components**
- Defines **control flow**
- **Guides agent design and engineering**

Modular Thinking in Agents

- Agents are composed of modules:
 - **Perception**
 - **Decision-making**
 - **Planning**
 - **Learning**
 - **Action**

Historical Evolution of Agent Architectures

- 1950s–1985: **symbolic reasoning agents**
- 1985–1995: **reactive agents**
- 1995–2010: **hybrid architectures**
- 2023–present: **LLM agents**

Deductive Reasoning Agents

- Agents based on **symbolic AI**
- **Knowledge base** representation
 - Internal state represented with logic
- **Logical inference and theorem proving**
 - Decision via logical inference



Limitations of Symbolic Agents

- **Computational complexity**
- **Slow decision making**
- **Difficulty handling dynamic environments**

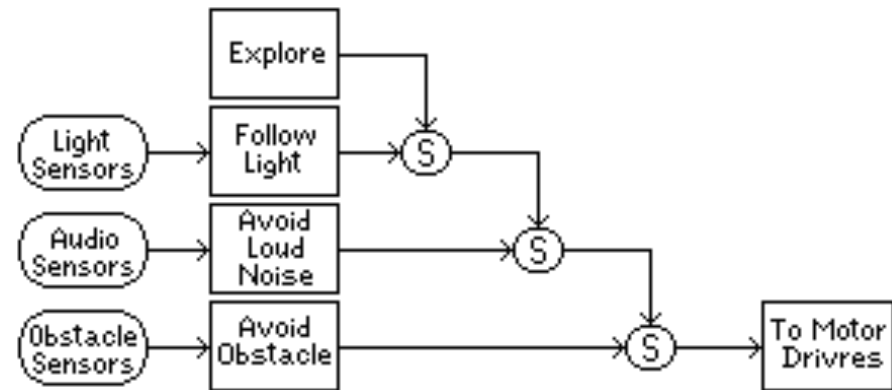
Reactive Agents

- Direct mapping: perception $\rightarrow$ action
- No complex reasoning
- Designed for fast response

```
function decide(perception)
  current_state <- INTERPRET-INPUT(perception)
  rule <- RULE-MATCH(current_state, rules)
  action <- RULE-ACTION(rule)
  return action
```

Subsumption Architecture

- Proposed by Rodney Brooks
- Layered behavior modules
- Lower layers override higher layers



Advantages of Reactive Agents

- Fast decision making
- Robust in dynamic environments
- Useful for robotics

Limitations of Reactive Agents

- Limited planning ability
- No long-term reasoning
- Hard to engineer complex behaviors

Deliberative Agents

- Agents reason about **goals**
- Use **symbolic world models**
- **Planning-based decision making**



Intentional Systems

- **Intentional stance**
 - **Michael E. Bratman.** *Intention, Plans and Practical Reason.* Harvard University Press, 1987.
- Agents described using **mental states**
- **Beliefs, desires, intentions**



Practical reasoning

Practical Reasoning = Deliberation + Means-Ends Reasoning

- ***Deliberation***: deciding what state of affairs an agent wants to achieve from (possibly conflicting) desires
- ***Means-Ends Reasoning***: deciding how an agent wants to achieve these states of affairs



BDI Architecture

- **Beliefs:** information about world
- **Desires:** goals
- **Intentions:** committed plans

BDI Control Loop

- Decision-making is a loop:
 - **Observe environment**
 - **Update beliefs**
 - **Generate options**
 - **Select intentions**
 - **Plan and execute actions**

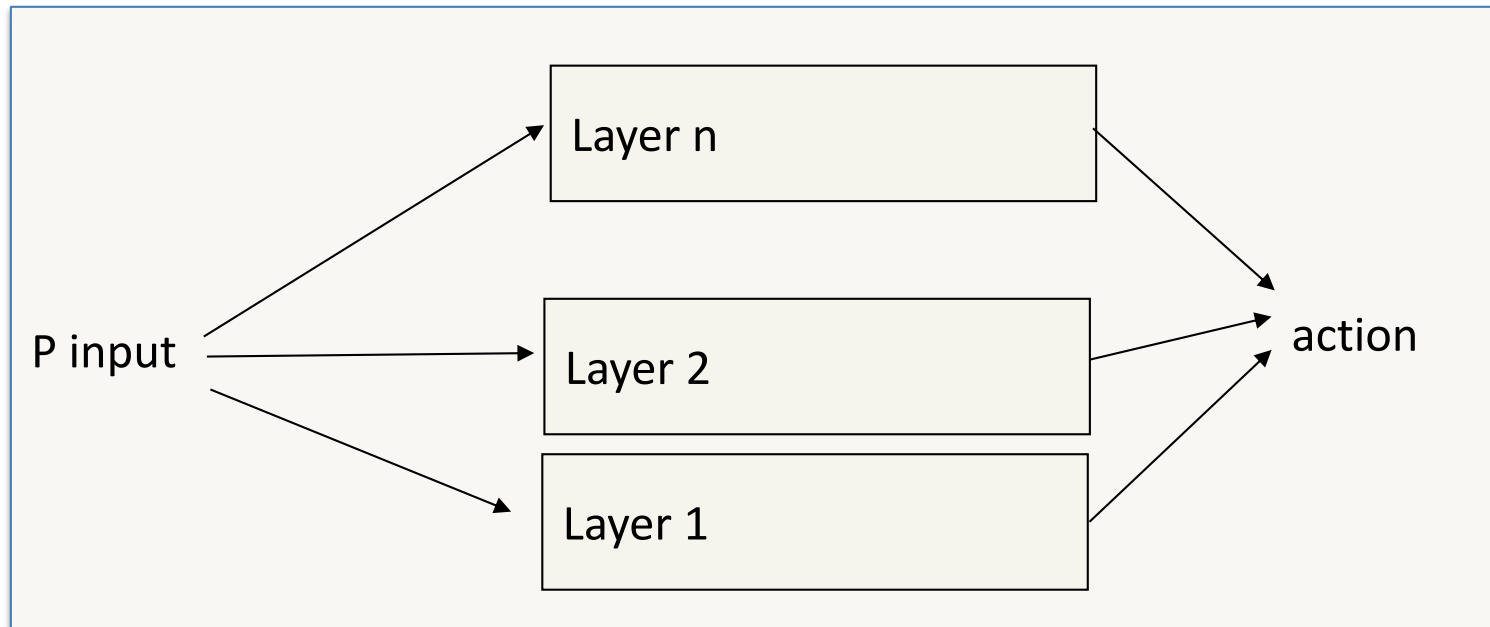
Limitations of Deliberative Agents

- Planning can be slow
- Incomplete information
- Hard real-world grounding

Hybrid Architectures

- Combine **reactive** and **deliberative** approaches
- Balance **speed** and **intelligence**

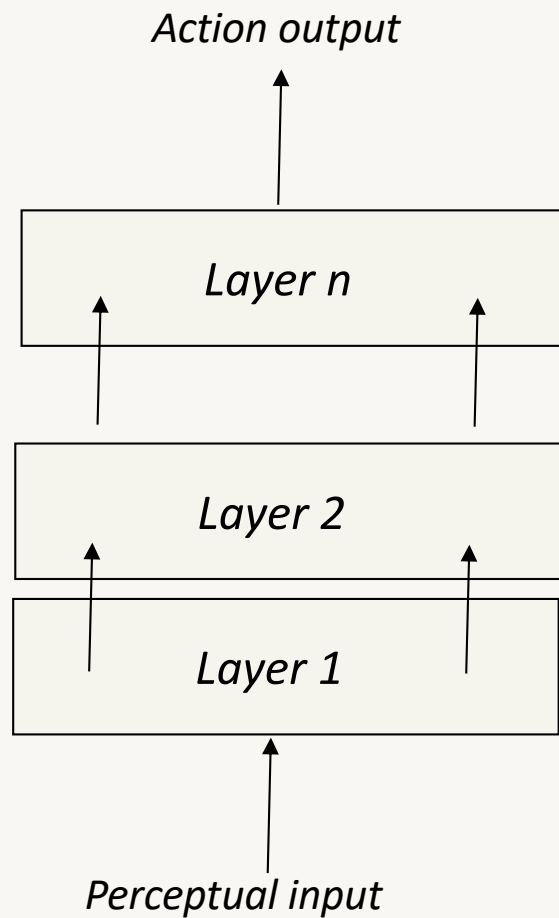
Hybrid architectures: horizontal layering



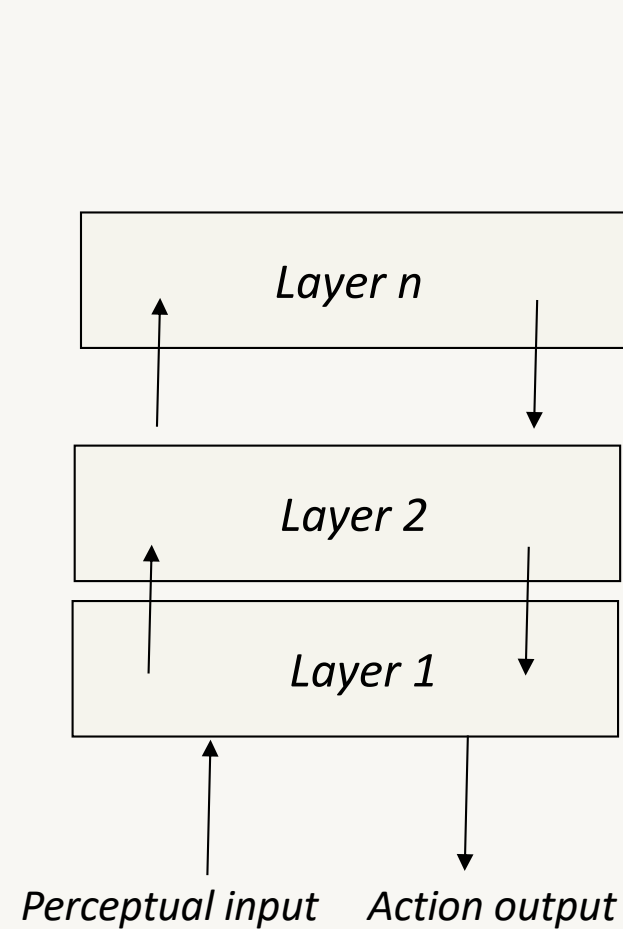
- **Advantages:** simple, distributed, fault-tolerant
- **Disadvantages:**
 - global behavior may not be coherent
 - difficult to build

Hybrid architectures: vertical layering

Vertical (1 pass control)

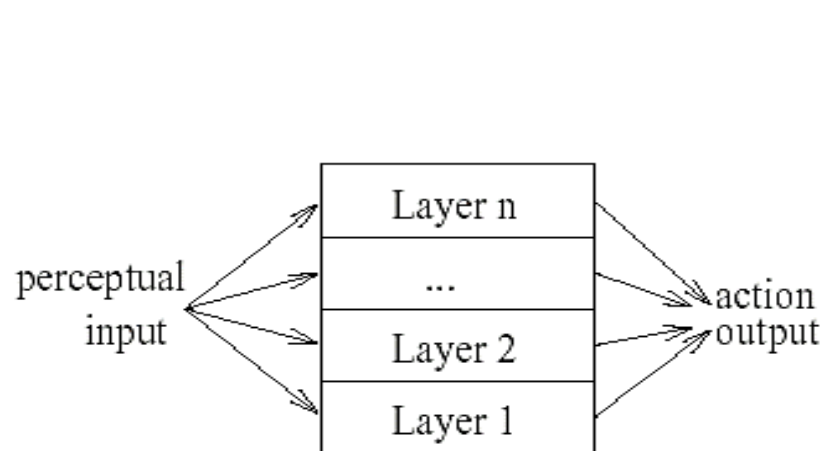


Vertical (2 passes control)

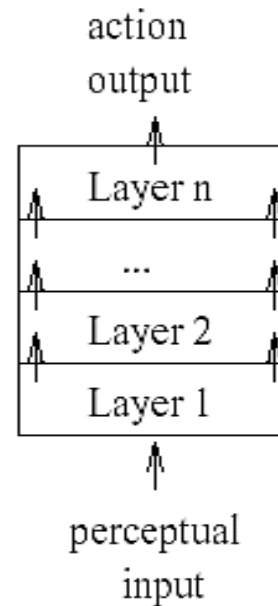


Comparing hybrid architectures

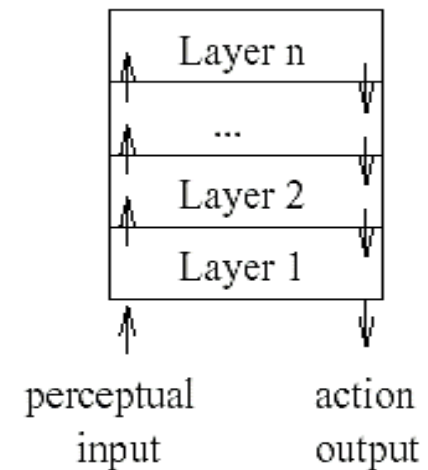
m possible actions suggested by each layer, n layers



(a) Horizontal layering



(b) Vertical layering
(One pass control)



(c) Vertical layering
(Two pass control)

**complex decision making
regarding the action output**

**not tolerant to
layer failure**

DARPA grand challenge

Main goal of the challenge:

- develop an autonomous car (i.e., self-driving car):
 - capable of **traversing unrehearsed off-road terrain**
 - travel a **175 mile** long course through the **Mojave desert**
 - **take no more than 10 hours**

DARPA grand challenge

- Teams
 - 2004 (1M\$ prize): 107 teams (15 finalists, 0 finished)
 - 2005 (2M\$ prize): 195 teams (23 finalists, 5 finished)
- **The route is kept secret** until 2h before the race
 - At this time they received a route description in RDDF format

Stanley – 2005 Winner



Volkswagen R5 (4WD)

- treats autonomous navigation as a software problem

Stanley: sensory equipment



Perception: roof rack that houses

- 5 SICK laser range finders
- camera for long range perception
- 2 RADAR sensors
- antennae: 1 for GPS, 1 for GPS compass, and 1 radio antennae

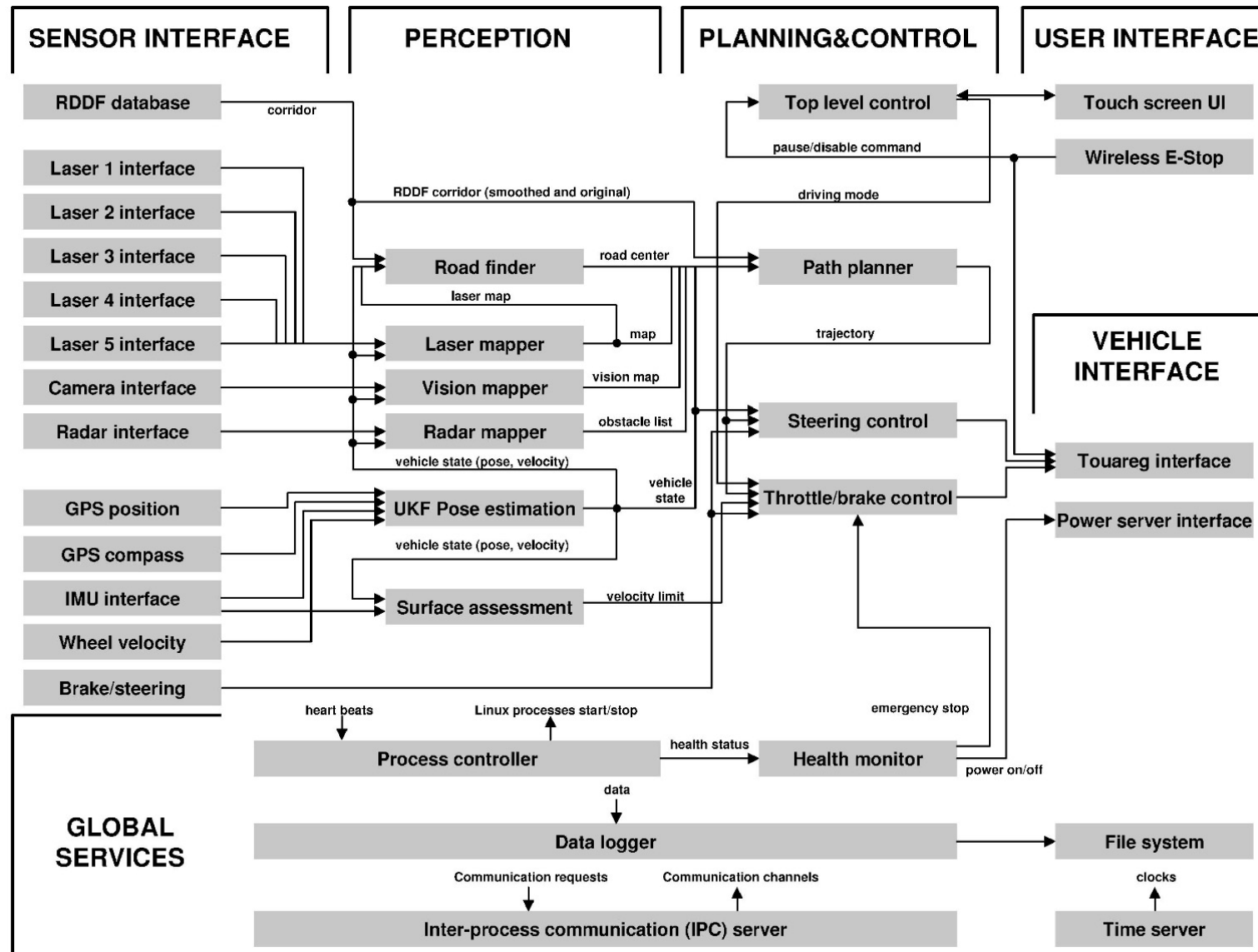
Stanley control



3 main actuators

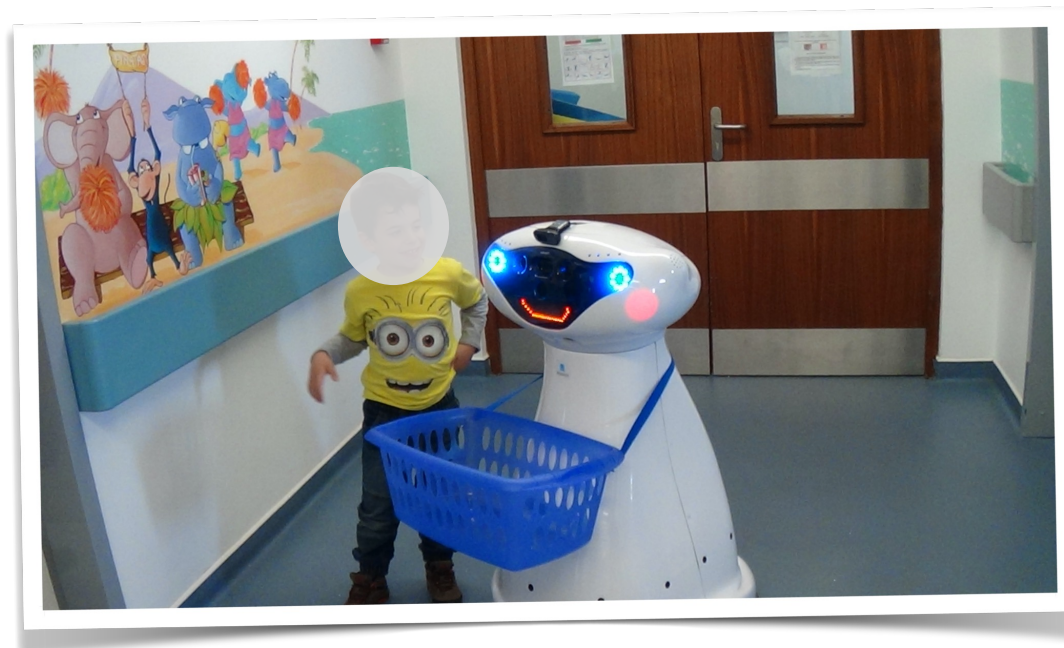
- brakes
- throttle
- steering

Stanley architecture



INSIDE Project

- A **key goal** of the INESIDE project:
- Develop a networked robot system that can help kids with ASD within therapy sessions



INSIDE Project

- **Main features** of the robot:
 - Fully autonomous and has a key role in the therapy sessions
 - Child-robot interaction should be unconstrained as possible (e.g, children could freely move around the world)
 - Child performs several fun activities in the therapy sessions



INSIDE Project

- Partners



INSIDE Project

- Venue: Hospital Garcia da Horta



INSIDE Project

- Key design choices for the interaction scenarios:
 - Child interacts with the robot during the **therapy session**
 - Following the therapist's guidance, the child engages in **game activities** with robot
 - Games are designed to address **specific therapeutic needs**

INSIDE Project

- Task 1: Addressing the social skill of **responding to a request**
 - Robot requests the child to search for colored balls in the room
 - The child should search the balls, retrieve them, and **give them back to the robot**



INSIDE Project

- Task 2: Addressing **social reciprocity and empathy**
 - Both robot and child move across the room
 - The robot finds its path blocked, requesting assistance from the child



INSIDE Project

- Task 3: Addressing the social skill of **asking for help**
 - Child is given a puzzle to solve
 - Pieces are missing in the puzzle
 - The child must **ask the robot** information on the missing pieces



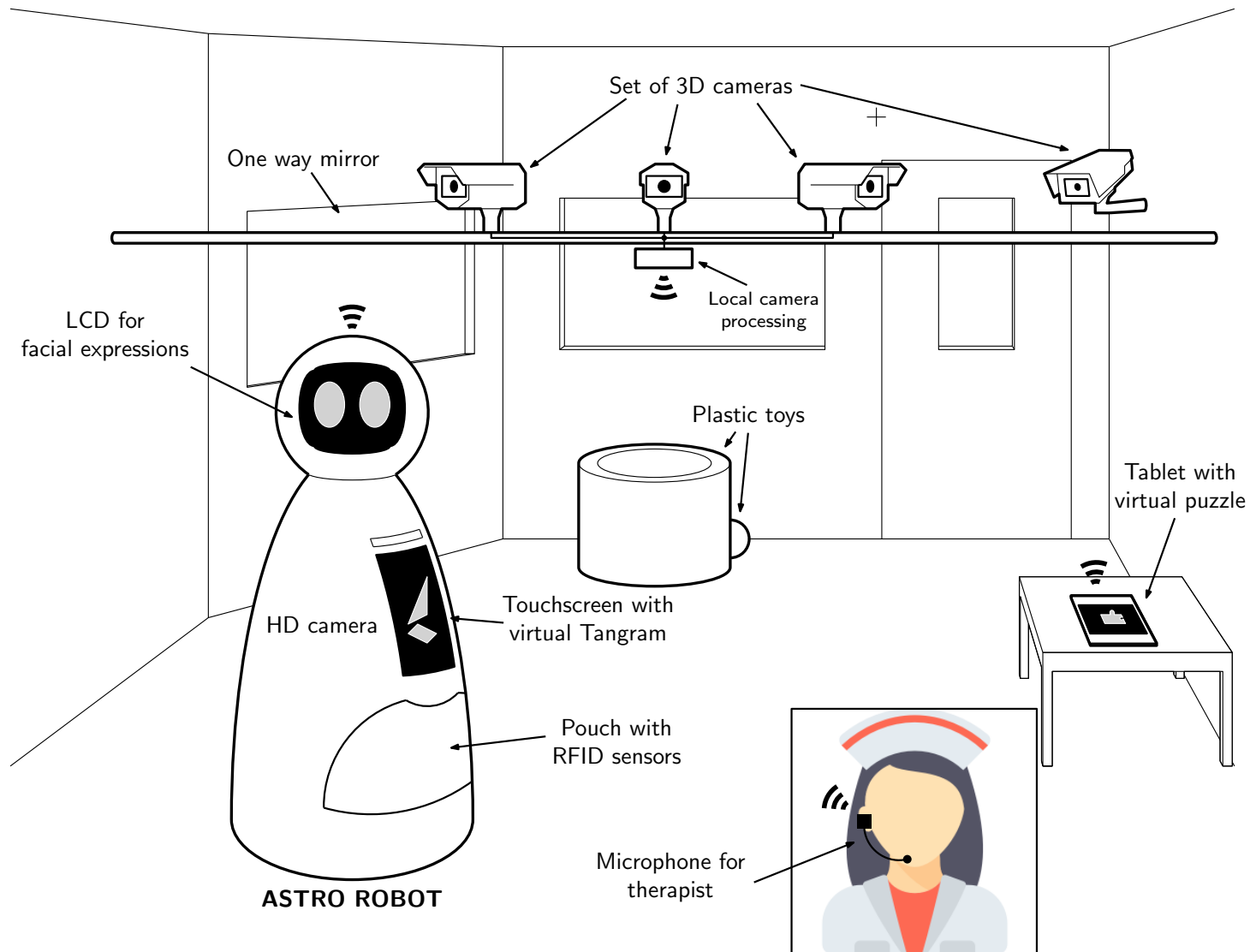
INSIDE Project

- Task 4: Addressing skill of **turn taking**
 - Child and robot are playing the interactive tangram game
 - Child and robot alternate in placing tangram pieces
 - Sometimes, the robot **asks for the child's assistance** in placing a piece



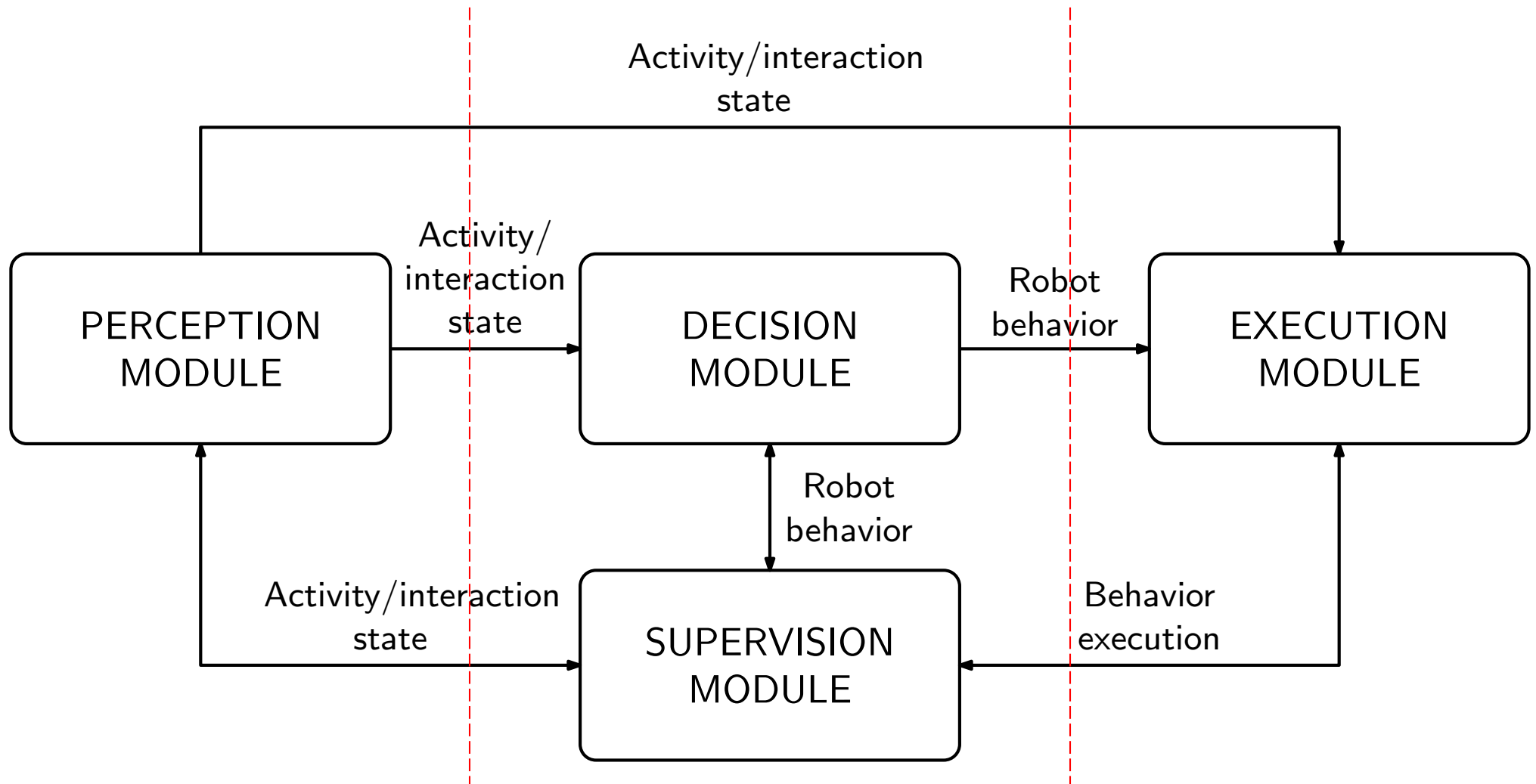
INSIDE Project

- The INSIDE Architecture

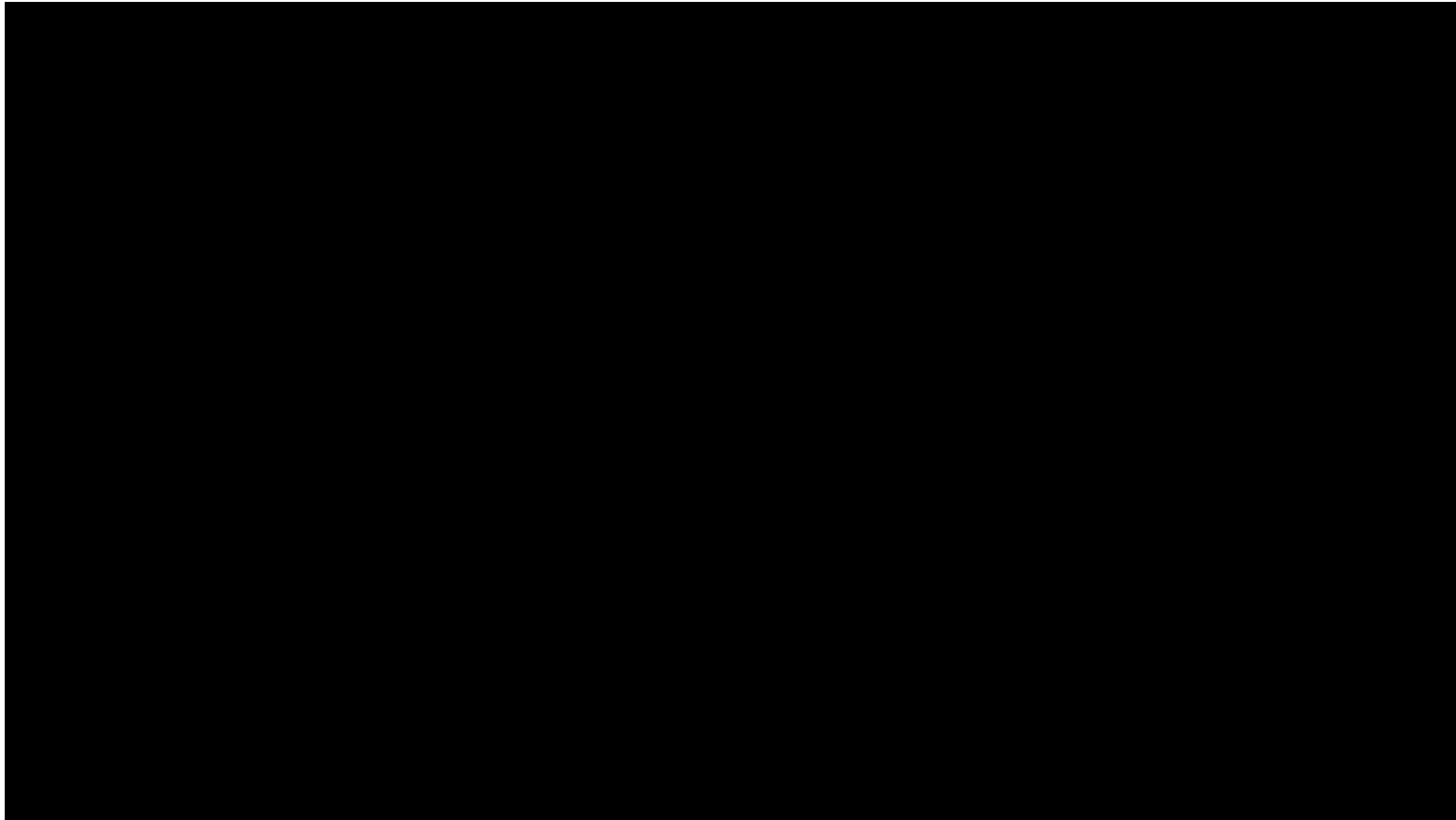


INSIDE Project

- The INSIDE Architecture



INSIDE Project



Return of Agents: Agentic AI

- **Large Language Models** enable new agents
- Agents interact with **tools, APIs, environments**

LLM Agent Architecture

- LLM reasoning engine
- Memory
- Tools/APIs
- Control loop

Agent Frameworks

- LangGraph
- AutoGen
- CrewAI
- n8n
- Model Context Protocol

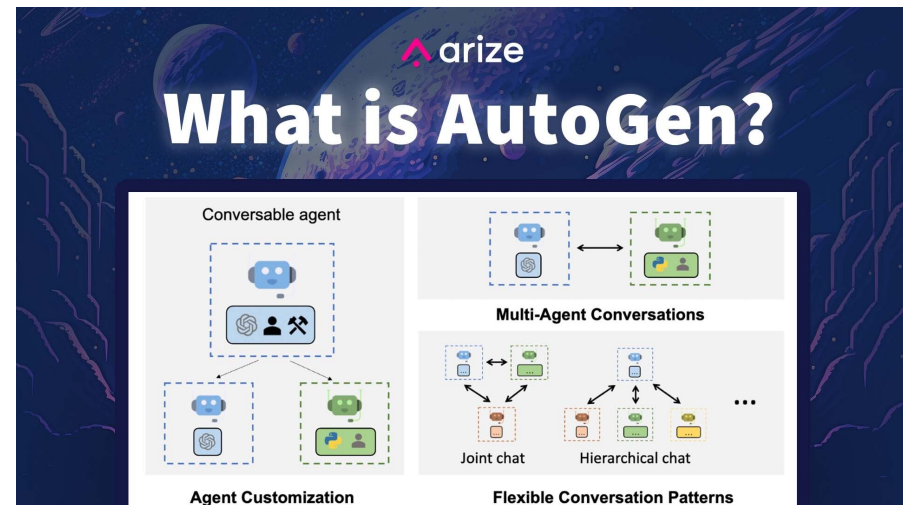
LangGraph

- Key Idea: **Graph-based agent workflows**
- **Strengths**
 - Deterministic control flow
 - Easy debugging
 - Production-ready workflows
- **Connection to MAS**
 - Similar to state machines / workflow agents



AutoGen

- Key Idea: **Conversational multi-agent systems**
- **Strengths**
 - Natural multi-agent collaboration
 - Good for reasoning tasks
- **Connection to MAS**
 - Similar to agent societies and negotiation systems



CrewAI

- Key Idea: **Role-based agent teams**

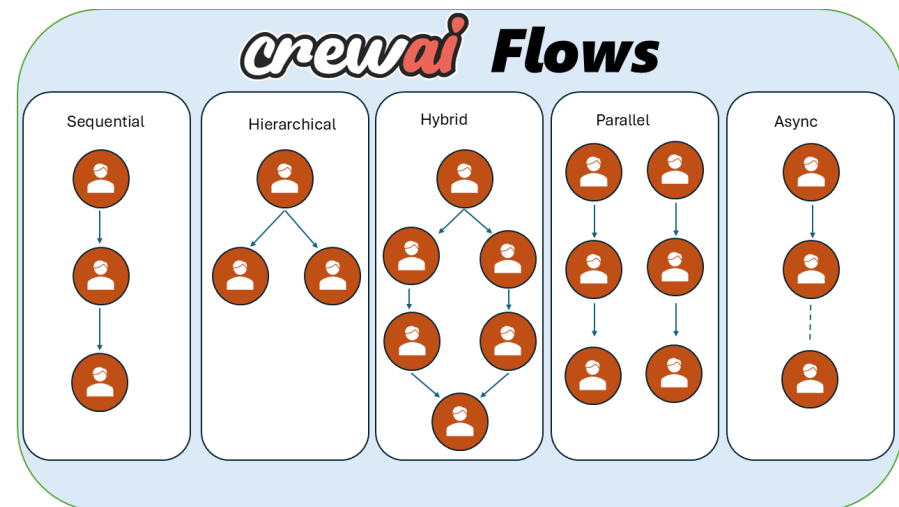
- Agents are organized as a crew

- **Strengths**

- Structured collaboration
 - Simple design for workflows

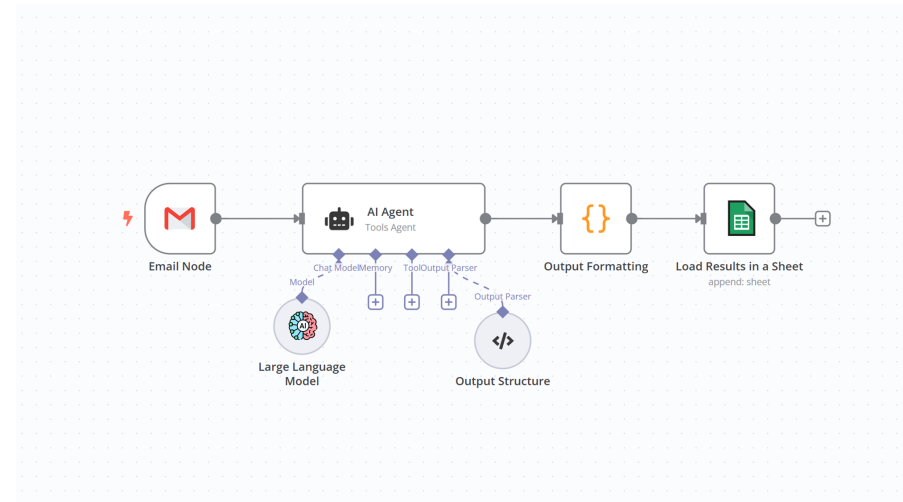
- **Connection to MAS**

- Similar to organizational multi-agent models



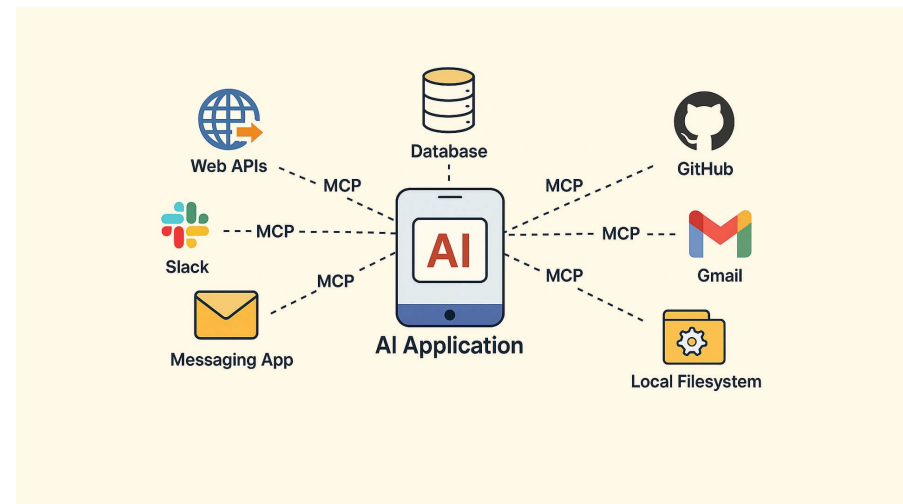


- Key Idea: **Workflow automation with LLM integration**
- Architecture: **Low-code automation workflows**
- **Strengths**
 - visual workflow design
 - integration with many APIs
- **Connection to MAS**
 - Similar to workflow-based agent systems.



Model Context Protocol

- Key Idea: **Standard interface between agents and tools**
- **Capabilities**
 - tool discovery
 - structured tool invocation
 - standardized APIs
- **Benefits**
 - interoperability between agents and services
 - easier tool integration
- **Connection to MAS**
 - Acts as an environment interface for agents



Challenges

- **planning reliability**
- **hallucinations**
- **agent coordination**
- **long-term memory**
- **cost of inference**
- **safety and alignment**

Thank You



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